U.G.C. NET Exa
COMPUTE
(Solved with Expla

1. A ripple counter is a (*n*):

(a) Synchronous Counter
(b) Asynchronous counter
(c) Parallel counter
(d) None of the above

Ans : (b) A **ripple counter** is an **asynchronous:** counter where only the first flip-flop is clocked by an external clock. All subsequent flip-flops are clocked by the output of the preceding flip-flop. **Asynchronous counters** are also called ripple-counters because of the way the clock pulse ripples its way through the flip-flops.

2. 8085 microprocessor has _______ bit ALU.

(a) 32 (b) 16
(c) 8 (d) 4

Ans : (c) 8085 microprocessor has 8 bit ALU.

3. The register that stores the bits required to mask the interrupts is

(a) Status register
(b) Interrupt service register
(c) Interrupt mask register
(d) Interrupt request register

Ans : (c) Interrupt Mask Register (IMR): This register stores the bits required to mask the interrupt inputs.

4. Which of the following in 8085 microprocessor performs
HL = HL + HL ?

(a) DAD D (b) DAD H
(c) DAD B (d) DAD SP

Ans : (b) DAD H: double add HL to HL $\Rightarrow$ HL = HL + HL

5. In ________ addressing mode, the operands are stored in the memory. The address of the corresponding memory location is given in a register which is specified in the instruction.

(a) Register direct (b) Register indirect
(c) Base indexed (d) Displacement

Ans : (b) Register indirect addressing means that the location of an operand is held in a register. It is also called indexed addressing or base addressing.

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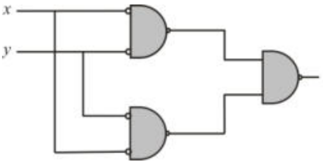
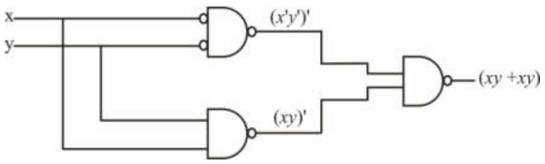
6. The output of the following circuit



(a) $X \cdot Y$ (b) $X + Y$

(c) $X \oplus Y$ (d) $\overline{X \oplus Y}$

Ans : (d) Given circuit is:



function $f = \overline{\overline{(\bar{x}.\bar{y})}.\overline{(x.y)}}$

$f = \overline{\overline{(\bar{x}.\bar{y})}}+\overline{\overline{(x.y)}}$

$f = (\bar{x}.\bar{y})+(x.y)$

$f = x + y = \overline{x \oplus y}$

7. Which of the following statements is/are True regarding some advantages that an object-oriented DBMS (OODBMS) offers over a relational database?

I. An OODBMS avoids the "impedance mismatch" problem.

II. An OODBMS avoids the "phantom" problem.

III. An OODBMS provides higher performance concurrency control than most relational database.

IV. An OODBMS provides faster access to individual data objects once they have been read from disk.

(a) II and III only (b) I and IV only

(c) I, II, and III only (d) I, III and IV only

Ans : (b) I. An OODBMS avoids the "impedance mismatch" problem. This is true.

II. An OODBMS doesn't avoids the "Phantom" problem.

III. An OODBMS doesn't provides higher performance concurrency control than most relational database since it is distributed.

IV. An OODBMS provides faster access to individual data objects once they have been read from disk. This is true.

8. **The Global conceptual Schema in a distributed database contains information about global relations. The condition that all the data of the global relation must be mapped into the fragments, that is, it must not happen that a data item which belongs to a global relation does not belong to any fragment, is called:**
(a) Disjointness condition
(b) Completeness condition
(c) Reconstruction condition
(d) Aggregation condition

Ans : (b) Completeness Condition: All the data of the global relation must be mapped into the fragments; it must not happen that a data item which belongs to a global relation does not belong to any fragment.

9. **Suppose database table T1 (P, R) currently has tuples {(10, 5), (15, 8), (25, 6)} and table T2 (A, C) currently has {(10, 6), (25, 3), (10, 5)}. Consider the following three relational algebra queries RA1, RA2 and RA3.**
RA1: T1 $\bowtie_{T1.P=T2.A}$ T2 where $\bowtie$ is natural join symbol.
RA2:T1 $\bowtie_{T1.P=T2.A}$ T2 where $\bowtie$ is left outer join symbol.
RA3: T1 $\bowtie_{T1.P=T2.A \text{ and } T1.R= T2.c}$ T2
The number of tuples in the resulting table of RA1, RA2 and RA3 are given by:
(a) 2, 4, 2 respectively
(b) 2, 3, 2 respectively
(c) 3, 3, 1 respectively
(d) 3, 4, 1 respectively

Ans : (d) RA1: T1 $\bowtie_{T1.P=T2.A}$ T2 where $\bowtie$ is natural join symbol. It will result 3 tuples:

P = A	R	C
10	5	6
10	5	5
25	6	3

RA2:T1 $\bowtie_{T1.P=T2.A}$ T2 where $\bowtie$ is left outer join symbol. It will result in 4 tuples.

P = A	R	C
10	5	6
10	5	5
15	8	null
25	6	3

RA3: T1 $\bowtie_{T1.P=T2.A \text{ and } T1.R= T2.c}$ T2. It will result in 1 tuple.

P = A	R = C
10	5

10. **Consider the table R with attributes A, B and C. The functional dependencies that hold on R**

are: $A \to B.C \to AB.$ Which of the following statements is/are True?

I. The decomposition of R into R_1(C, A) and R_2(A, B) is lossless.

II. The decomposition of R into R_1(A, B) and R_2(B, C) is lossy.

(a) Only I (b) Only II

(c) Both I and II (d) Neither I nor II

Ans : (c) Statement - 1

R_1(C, A) R_1 (A, B)

$R_1 \cap R_2$ = A which is superkey for A

So its lossless Join

Statement -2

R_1(A, B) R_1 (B, C)

$R_1 \cap R_2$ = B which is not superkey for any relations so its lossy Join

Both are correct option.

11. Consider the following ORACLE relations:

One (*x, y*) = {<2, 5>, <1, 6>, <1, 6>, <1, 6>, <4, 8>, <4, 8>}

Two (*x, y*) = {<2, 55>, <1, 1>, <4, 4>, <1, 6>, <4, 8>, <4, 8>}

Consider the following two SQL queries SQ1 and SQ2:

SQ1: SELECT * FROM One)
EXCEPT
(SELECT * FROM Two);

SQ2: SELECT * FROM One)
EXCEPT ALL
(SELECT * FROM Two);

For each of the SQL queries, what is the cardinality (number of rows) of the result obtained when applied to the instances above?

(a) 2 and 1 respectively

(b) 1 and 2 respectively

(c) 2 and 2 respectively

(d) 1 and 1 respectively

Ans : (b) Except clause returns all records present in the first relation and absent in the second relation. But this clause eliminates duplicates tuples.

SQ1: SELECT * FROM One)
EXCEPT
(SELECT * FROM Two);

It will result into single tuple (2, 5) because all duplicate tuples will be removed.

Except all returns all records from the first table which are not present in the second table, leaving the duplicates tuples as it is.

SQ2: SELECT * FROM One)
EXCEPT ALL
(SELECT * FROM Two);

It will result into 2 tuples < (2, 5), (1, 6) > because it will not remove duplicate.

12. **Which one of the following pairs is correctly matched in the context of database design?**

	List-I (Database term)		**List-II (Definition)**
A.	Specialization	**1.**	Result of taking the union of two or more disjoint (lower-level) entity sets to produce a higher-level entity set.
B.	Generalization	**2.**	Express the number of entities to which another entity can be associated via a relationship set.
C.	Aggregation	**3.**	Result of taking a subset of a higher-level entity set to form a lower-level entity set.
D.	Mapping cardinalities	**4.**	An abstraction in which relationship sets (along with their associated entity sets) are treated as higher-level entity sets, and can participate in relationships.

Codes:

	A	**B**	**C**	**D**
(a)	4	1	2	3
(b)	4	3	2	1
(c)	3	4	1	2
(d)	2	1	4	2

Ans : (d) (i) Specialization: is uses top to down approach and it is opposite to Generalization.
(ii) Generalization: is a bottom-up approach in which two or more entities can be generalized to a higher level entity if they have some attributes in common.
(iii) Aggregation: It is treated as higher level entity sets and can participate in relationships.
(iv) Mapping cardinalities: Number of entities to which another entity can be associated via a relationship set.

13. Consider a raster grid having *XY*-axes in positive *X*-direction and positive upward *Y*-direction with X_{max} = 10, X_{min} = -5, Y_{max} = 11, and Y_{min} = 6. What is the address of memory pixel with location (5, 4) in raster grid assuming base address 1 (one)?

(a) 150 (b) 151
(c) 160 (d) 161

Ans : (d)
addr(X, Y) = addr(0, 0) + Y rows × (X_m + 1) + Y(All in bytes)
A(5, 4) = 1 + (5(-5)) × 6 + (4 + 6)

14. Consider a *N*-bit plane frame buffer with *W*-bit wide lookup table with *W* > *N*. How many intensity levels are available at a time?

(a) 2^N (b) 2^W
(c) 2^{N+W} (d) 2^{N-1}

Ans : (a) *N*-bit colour frame buffer:–
Colour or gray scales are incorporated into a frame buffer rater graphics device by using additional bit planes. The intensity of each pixel on the CRT is controlled by a corresponding pixel location in each of the *N*-bit planes. The binary value from each of the *N*-bit planes is located into corresponding positions in a register. The resulting binary number is interpreted as an intensity level between 0 (dark) and 2^n - 1 (full intensity). This is converted into an analog voltage between 0 and the maximum voltage of the electron gun by the DAC, **A total of 2^n intensity levels are possible.**

15. Consider the Breshenham's line generation algorithm for a line with gradient greater than one, current point (x_i, y_i) and decision parameter, d_i. The next point to be plotted (x_{i+1}, y_{i+1}) and updated decision parameter, d_{i+1}, for d_i < 0 are given as _________ .

(a) $x_{i+1} = x_i + 1$
$y_{i+1} = y_i$
$d_{i+1} = d_i + 2dy$

(b) $x_{i+1} = x_i$
$y_{i+1} = y_i + 1$
$d_{i+1} = d_i + 2dx$

(c) $x_{i+1} = x_i$
$y_{i+1} = y_i + 1$
$d_{i+1} = d_i + 2(dx - dy)$

(d) $x_{i+1} = x_i + 1$
$y_{i+1} = y_i + 1$
$d_{i+1} = d_i + 2(dy - dx)$

Ans : (b) According to **Breshenham's** line generation algorithm:

$x_{i+1} = x_i$
$y_{i+1} = y_i + 1$
$d_{i+1} = d_i + 2dx$

16. A point P(2, 5) is rotated about a pivot point (1, 2) by 60^0. What is the new transformed point P'?

(a) (1, 4) (b) (-1, 4)
(c) (1, -4) (d) (-4, 1)

Ans:(b) $= \begin{bmatrix} \cos\theta & -\sin\theta & -t_x\cos\theta+t_y\sin\theta+t_x \\ \sin\theta & \cos\theta & -t_x\sin\theta-t_y\cos\theta+t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$

Given, θ = 60 and x = 2, y = 5 then

$$= \begin{bmatrix} 0.5 & -0.866 & -1\times0.5+2\times0.866+1 \\ 0.866 & 0.5 & -1\times0.866-2\times0.5+2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & -0.866 & 2.232 \\ 0.866 & 0.5 & 0.134 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} -1.098 \\ 4.366 \\ 1 \end{bmatrix}$$

Therefore, P' = (-1, 4)

17. In perspective projection (from 3D to 2D), objects behind the centre of projection are projected upside down and backward onto the view-plane. This is known as ________ .

(a) Topological distortion
(b) Vanishing point
(c) View confusion
(d) Perspective foreshortening

Ans : (c) View confusion: Projects behind the center of the projection are projected upside down and backward on to the view plane.

18. The Liang-Barsky line clipping algorithm uses the parametric equation of a line from (x_1, y_1) to (x_2, y_2) along with its infinite extension which is given as:

$$\mathbf{x = x_1 + \Delta x.u}$$
$$\mathbf{y = y_1 + \Delta y.u}$$

Where Δx = x_2 - x_1, Δy = y_2 - y_1 and u is the parameter with $0 \le u \le 1$. A line AB with end points A(-1, 7) and B(11, 1) is to be clipped against a rectangular window with x_{min} = 1, x_{max} = 9, y_{min} = 2, and y_{max} = 8. The lower and upper bound values of the parameter u for the clipped line using Liang-Barsky algorithm is given as:

(a) $\left(0, \frac{2}{3}\right)$ (b) $\left(\frac{1}{6}, \frac{5}{6}\right)$

(c) $\left(0, \frac{1}{3}\right)$ (d) (0, 1)

Ans : (b) first we will calculate Δx and Δy:

ie $\Delta x = x_2 - x_1 = 11 - (-1) = 11 + 1 = 12$.

$\Delta y = y_2 - y_1 = 1 - 7 = -6$

Now $P_1 = -\Delta x = -12$

$P_2 = \Delta x = 12$

$P_3 = -\Delta y = 6$

$P_4 = \Delta y = -6$

$Q_1 = x_1 - x_{min} = -1 - 1 = -2$

$Q_2 = x_{max} - x_1 = 9 - (-1) = 9 + 1 = 10$

$Q_3 = y_1 - y_{min} = 7 - 2 = 5$

$Q_4 = y_{max} - y_1 = 8 - 7 = 1$

$P_1, P_4 < 0$ and $P_2, P_3 > 0$

Initially: $t_1 = 0$, $t_2 = 1$

$t_1 = \max(0, Q_1/P_1, Q_4/P_4)$

$= \max(0, 2/12, 1/-6)$

$= 1/6$

$t_2 = \min(1, Q1/P_1, Q_4/P_4)$

$= \min(1, 10/12, 5/6)$

$= 5/6$

i.e. $\cup$ ranges between (1/6, 5/6)

19. Match the following with reference to Functional programming history:

	List-I		List-II
A.	Lambda calculus	**1.**	Church, 1932
B.	Lambda calculus as programming language	**2.**	Wordsworth, 1970
C.	Lazy evaluation	**3.**	Haskel, 1990
D.	Type classes	**4.**	Mecarthy, 1960

Codes:

	A	B	C	D
(a)	4	1	3	2
(b)	1	4	2	3
(c)	3	2	4	1
(d)	2	1	4	3

Ans : (b) (i) Lambda calculus: is a formal system in mathematical logic for expressing computation based on function abstraction and application using variable binding and substitution. The lambda calculus was introduced by mathematician Alonzo church in the 1930's as part of an investigation into the "foundation of mathematics (1932)".

Until the 1960's when its relation to programming languages was clarified, the λ-calculus was only a formalism. To LISP by MeCarthy.
(ii) Lazy evaluation: Was introduced in 1970 by wordsworth.
(iii) Type classes: Was introduced in 1990 by Haskel.

20. Aliasing in the context of programming languages refers to:
(a) Multiple variables having the same location
(b) Multiple variables having the same identifier
(c) Multiple variables having the same value
(d) Multiple use of same variable

Ans : (a) Aliasing: Describes a situation in which a data location in memory can be accessed through different symbolic names in the program. Thus, modifying the data through one name implicitly modifies the value associated with all aliased names, which may not be expected by the programmer.

21. Assume that the program 'P' is implementing parameter passing with 'call by reference'. What will be printed by following print statements in P?

```
Program P( )
{
    x = 10;
    y = 3;
    funb (y, x, x); print x; print y;
}
funb (x, y, z)
{
    y = y + 4;
    z = x + y + z;
}
```

(a) 10, 7 (b) 31, 3
(c) 10, 3 (d) 31, 7

Ans : (b) Inside funb ( ), x will actually refer to y of main ( ); and y and z will refer to x of main ( ).
The statement $y = y + 4$;
will result in 14 and statement.
$z = x + y + z$ will make
$z = 3 + 14 + 14$
$= 31$
(because y and z point to same variable x of main).
Since z refers to x of main ( ), main will print 31. z will be stored back in x = 31 and y will remain as it is, because there was no change for it. Then y = 3.
Printed output is 31, 3.

22. The regular grammar for the language L = $\{a^n b^m \mid n + m$ is even$\}$ is given by
(a) $S \rightarrow S_1 \mid S_2$
$S_1 \rightarrow aS_1 \mid A_1$
$A_1 \rightarrow bA_1 \mid \lambda$
$S_2 \rightarrow aaS_2 \mid A_2$
$A_2 \rightarrow bA_2 \mid \lambda$

(b) $S \to S_1 \mid S_2$

$S_1 \to aS_1 \mid aA_1$

$S_2 \to aaS_2 \mid A_2$

$A_1 \to bA_1 \mid \lambda$

$A_2 \to bA_2 \mid \lambda$

(c) $S \to S_1 \mid S_2$

$S_1 \to aaaS_1 \mid aA_1$

$S_2 \to aaS_2 \mid A_2$

$A_1 \to bA_1 \mid \lambda$

$A_2 \to bA_2 \mid \lambda$

(d) $S \to S_1 \mid S_2$

$S_1 \to aaS_1 \mid A_1$

$S_2 \to aaS_2 \mid aA_2$

$A_1 \to bbA_1 \mid \lambda$

$A_2 \to bbA_2 \mid \lambda$

Ans : (d) Given, $L = \{a^n b^m \mid (n + m) \text{ is even}\}$

For $(n + m)$ to be even either n and m both are even or n and m both are odd.

So, for n and m to be even, grammar is:

$S_1 \to aaS_1 \mid A_1$

$A_1 \to bbA_1 \mid \in$

For n and m odd, grammar is:

$S_2 \to aaS_2 \mid aA_2$

$A_2 \to bbA_2 \mid b$

Now, combine both; then resultant grammar is:

$S \to S_1 \mid S_2$

$S_1 \to aaS_1 \mid A_1$

$A_1 \to bbA_1 \mid \in$

$S_2 \to aaS_2 \mid aA_2$

$A_2 \to bbA_2 \mid b$

23. Let $\Sigma = \{a, b\}$ and language $L = \{aa, bb\}$. Then, the complement of L is

(a) $\{\lambda, a, b, ab, ba\} \cup \{w \in \{a, b\}^* \mid |w| > 3\}$

(b) $\{a, b, ab, ba\} \cup \{w \in \{a, b\}^* \mid |w| \geq 3\}$

(c) $\{w \in \{a, b\}^* \mid |w| > 3\} \cup \{a, b, ab, ba\}$

(d) $\{\lambda, a, b, ab, ba\} \cup \{w \in \{a, b\}^* \mid |w| \geq 3\}$

Ans : (d) String generated by language L = {aa, bb} will be even length.

So, complement will be universal set of strings– {aa, bb}. i.e. $\{\lambda, a, b, ab, ba\}$ and strings $\{w \in \{a, b\}^* \mid |w| \geq 3\}$,

i.e. string of length greater than or equal to 3.

24. Consider the following identities for regular expressions:

(i) $(r + s)^* = (s + r)^*$

(ii) $(r^*) = r^*$

(iii) $(r^* s^*) = (r + s)^*$

Which of the above identities are true?

(a) (i) and (ii) only
(b) (ii) and (iii) only
(c) (iii) and (i) only
(d) (i), (ii) and (iii)

Ans : (d) $\Rightarrow$ $(r + s)^*$ will generate any string containing r or s or both. We can draw DFA for $(r + s)^*$ and it is same as $(s + r)^*$. It is a regular expression.

$\Rightarrow$ $(r^*)^*$ will generate any string containing r and its DFA can be draw easily and it is same as r^*. It is also a regular expression.

$\Rightarrow$ $(r^* s^*)^*$ will generate any string containing r or s or both. We can draw DFA for $(r^* s^*)^*$ and it is same as $(r + s)^*$. It is a regular expression. All option are true.

25. Suppose transmission rate of a channel is 32 kbps. If there are '8' routes from source to destination and each packet p contains 8000 bits. Total end to end delay in sending packet *P* is _________ .

(a) 2 sec (b) 3 sec
(c) 4 sec (d) 1 sec

Ans : (a) Each router takes

$$=\frac{8000}{32000} \text{ second}$$

All 8 routes takes

$$=8\times\left(\frac{8000}{32000}\right)=2 \text{ second}$$

26. Consider the following statements:

I. High speed Ethernet works on optic fiber.

II. A point to point protocol over Ethernet is a network protocol for encapsulating PPP frames inside Ethernet frames.

III. High speed Ethernet does not work on optic fiber.

IV. A point to point protocol over Ethernet is a network protocol for encapsulating Ethernet frames inside PPP frames.

Which of the following is correct?

(a) I and II are true; III and IV are false.
(b) I and II are false; III and IV are true.
(c) I, II, III and IV are true.
(d) I, II, III and IV are false.

Ans : (a) (A). High speed Ethernet works on optic fiber. **(B).** A point to point protocol over Ethernet is a network protocol for encapsulating frames inside Ethernet frames not for encapsulating Ethernet frames inside PPP frames.

27. In CRC checksum method, assume that given frame for transmission is 11010111011 and the generator polynomial is

$$G(x) = x^4 + x + 1.$$

After implementing CRC encoder, the encoded word sent from sender side is ____ .

(a) 11010110111110
(b) 11101101011011
(c) 110101111100111
(d) 110101111001111

Ans : (a) Given, message is: 1101011011 and generator polynomial is $G(x) = x^4 + x + 1$, which is equivalent to 10011. Now we add (n - 1) 0's to the LSB of message, if CRC generator is of n-bits.
Therefore, message will be 11010110110000. Now we divide it by 10011, we will get 1110 as remainder, therefore transmitted message will be:
11010110111110.

28. A slotted ALOHA network transmits 200 bits frames using a shared channel with 200 kbps bandwidth. If the system (all stations put together) produces 1000 frames per second, then the throughput of the system is _______ .

(a) 0.268 (b) 0.468
(c) 0.368 (d) 0.568

Ans : (c) For pure aloha throughput (S) = Ge^{-2G}
For slotted aloha throughput (S) = Ge^{-G}
(it is maximum at G = 1).
i.e. S = 1* e^{-1}
S = 1/e
S = 1/2.71
S = 0.368

29. An analog signal has a bit rate of 8000 bps and a baud rate of 1000. Then analog signal has ______ signal elements and carry _____ data elements in each signal.

(a) 256, 8 bits (b) 128, 4 bits
(c) 256, 4 bits (d) 128, 8 bits

Ans : (a)

$$S = N \times \frac{1}{r}$$

$$\Rightarrow \quad r = \frac{N}{S} = \frac{8000}{1000} = 8 \text{ bits/bond}$$

Now,

$$\Rightarrow \quad r = \log_2 L$$

$$\Rightarrow \quad L = 2^r = 2^8 = 256$$

Therefore, each signal carry 8 bits which has 256 signals-elements

30. The plain text message BAHI encrypted with RSA algorithm using e = 3, d = 7 and n = 33 and the characters of the message are encoded using the values 00 to 25 for letters A to Z. Suppose character by character encryption was implemented. Then, the Cipher Text message is

(a) ABHI (b) HAQC
(c) IHBA (d) BHQC

Ans: (b) Using RSA algorithm:

If we want to get the ciphertext of BAHI using RSA algorithm, we will get:

(i) P^e mod n

(ii) C^d mod w

$B \rightarrow 2^3 = 8 \text{ mod } 33 = 8 = \text{H}$

$A \rightarrow 1^3 = 1 \text{ mod } 33 = 1 = \text{A}$

$H \rightarrow 8^3 = 17 \text{ mod } 33 = 17 = \text{Q}$

$I \rightarrow 9^3 = 3 \text{ mod } 33 = 3 = \text{C}$

So, the; ciphertent will be HAQC.

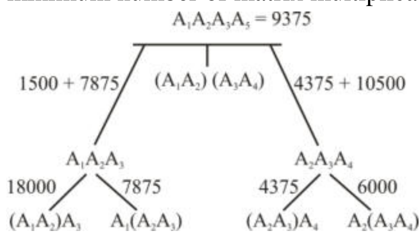
31. Consider the problem of a claim <A_1, A_2, A_3, A_4> of four matrices. Suppose that the dimensions of the matrices A_1, A_2, A_3, and A_4 are 30 × 35, 35 × 15, 15 × 5 and 5 × 10 respectively. The minimum number of scalar multiplications needed to compute the product $A_1 A_2 A_3 A_4$ is ________ .

(a) 14875 (b) 21000

(c) 9375 (d) 11875

Ans : (c) Given, dimensions of the matrices A_1, A_2, A_3 and A_4 are 30 × 35, 35 × 15, 15 × 5 and 5 × 10 respectively.

We have five possible orders to perform this matrix multiplication. Now we have to find the one which will have minimum number of matrix multiplication.



Therefore, minimum number of multiplications possible in order $(A_1(A_2A_3))\ A_4 = 9375$

32. Consider a hash table of size m = 10000, and the hash function $h(K)$ = floor (m(KA mod 1)) for $\text{A} = (\sqrt{5}-1)/2$. The key 123456 is mapped to location _________ .

(a) 46 (b) 41

(c) 43 (d) 48

Ans : (b) Given,

$$\text{h(k)} = \lfloor m(KA \text{ mod} 1) \rfloor$$

$$\text{h(123456)} = \lfloor 10000(123456 \times 0.61803 \text{ mod} 1) \rfloor$$

$$= \lfloor 10000(0.0041151) \rfloor$$

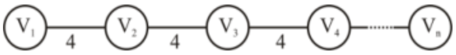
$$= \lfloor (41.151) \rfloor = 41$$

33. Consider a weighted complete graph G on the vertex set {v_1, v_2, ... v_n} such that the weight of the edge (v_i, v_j) is $4|i - j|$. The weight of minimum cost spanning tree of G is:

(a) $4n^2$ (b) n

(c) 4n - 4 (d) 2n - 2

Ans : (c) The minimum spanning tree of such a graph is:



weight of the minimum spanning trees

= 4 |2 - 1| + 4 |3 - 2| + 4 |4 - 3| + ... + 4 |*n* - (*n* - 1)|

= 4 + 4 + 4 + ... 4

= 4*n* - 4 = 4(*n* - 1)

34. A priority queue is implemented as a max-heap. Initially, it has five elements. The level-order traversal of the heap is as follows:

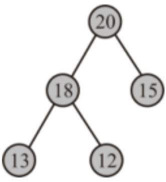
20, 18, 15, 13, 12

Two new elements '10' and '17' are inserted in the heap in that order. The level-order traversal of the heap after the insertion of the elements is:

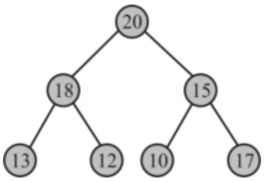
(a) 20, 18, 17, 15, 13, 12, 10
(b) 20, 18, 17, 12, 13, 10, 15
(c) 20, 18, 17, 10, 12, 13, 15
(d) 20, 18, 17, 13, 12, 10, 15

Ans : (d) Given level order of max-heap tree;

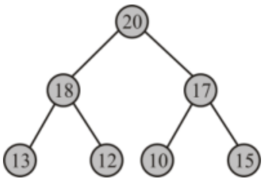
20, 18, 15, 13, 12



Now, 10 and 15 are inserted, tree will be:



Now there is a need to apply max-heapify. Hence,



level-order traversal is:

20, 18, 17, 13, 12, 10, 15

35. If there are n integers to sort, each integer has d digits, and each digit is in the set {1, 2, ... k}, radix sort can sort the numbers in:

(a) O(k(n + d)) (b) O(d(n + k))
(c) O((n + k)|gd) (d) O((n + d)|gk)

Ans : (b) Radix sort complexity is O(wn) for n keys which are integers of words size w. If there are n integers to sort, each integer has d digits, and each digit is in the set {1, 2,….,k} radix sort can sort the numbers in O(d(n + k)).

36. Match the following:

	List-I		List-II
A.	Prim's algorithm	**1.**	$O(V^2E)$
B.	Bellman-Ford algorithm	**2.**	O(VElgV)
C.	Floyd-Warshall algorithm	**3.**	O(ElgV)
D.	Johnson's algorithm	**4.**	$O(V^3)$

Where V is the set of nodes and E is the set of edges in the graph.

Codes:

	A	B	C	D
(a)	1	3	4	2
(b)	1	3	2	4
(c)	3	1	4	2
(d)	3	1	2	4

Ans : (c) Prim's algorithm to final minimum spanning tree in a weighted graph has time complexity O(E | log | V).

Bellman for algorithm to find single source shortest path in a weighted graph has time complexity $O(|V|\ |E|) = O(|V|^3)$.

Floyd-warshall algorithm to find all pair' shortest path in a graph has time complexity $O(|V|^3)$.

Johnson's algorithm to find all pair shortest path in a graph has time complexity $O(|V|^2 \log |V| + |V|\ |E|)$.

37. Constructors have _______ return type.

(a) void (b) char

(c) int (d) no

Ans : (d) A constructor resembles an instance method, but it differs form a method since it has no explicit return type, it is not implicity inherited and it usually has different rules for scope modifiers.

38. Method over-riding can be prevented by using final as a modifier at

(a) the start of the class.

(b) the start of method declaration.

(c) the start of derived class.

(d) the start of the method declaration in the derived class.

Ans : (b) To disallow a method from being overridden, specify final as a modifier at the start of its declaration. Methods declared as final cannot be overridden.

39. Which of the following is a correct statement?

(a) Composition is a strong type of association between two classes with full ownership.

(b) Composition is a strong type of association between two classes with partial ownership.
(c) Composition is a weak type of association between two classes with partial ownership.
(d) Composition is a weak type of association between two classes with strong ownership.

Ans : (a) Composition is a strong type of association between two classes with full ownership. In this relationship child objects does not have their lifecycle without parent object. If a parent object is deleted, all its child objects will also be deleted.

40. Which of the following is not a correct statement?
(a) Every class containing abstract method must be declared abstract.
(b) Abstract class can directly be initiated with 'new' operator.
(c) Abstract class can be initiated.
(d) Abstract class does not contain any definition of implementation.

Ans : (b) Abstract class cannot be directly initiated with new operator, since abstract class does not contain any definition of implementation it is not possible to create an abstract object.

41. Which of the following tag in HTML is used to surround information, such as signature of the person who created the page?
(a) <body> </body>
(b) <address> </address>
(c)
(d)

Ans : (b) The **<address> tag** is used to surround information, such as the signature of the person who created the page or the address of the address of the organization the page is about.
The **<body> tag** defines the document is body.
The **<em> tag** defines emphasized text.
The **<strong> tag** defines important text.

42. Java uses threads to enable the entire environment to be _________ .
(a) Symmetric (b) Asymmetric
(c) Synchronous (d) Asynchronous

Ans : (d) Java uses threads to enable the entire environment to be **asynchronous.** This helps in reducing inefficiency by preventing the waste of CPU cycles.

43. An Operating System (OS) crashes on the average once in 30 days, that is, the Mean Time Between Failures (MTBF) = 30 days. When this happens, it takes 10 minutes to recover the OS, that is, the Mean Tie To Repair (MTTR) = 10 minutes. The availability of the OS with these reliability figures is approximately:

(a) 96.97% (b) 97.97%
(c) 99.009% (d) 99.97%

Ans : (d) System crashes once in 30 days and need 10 minutes to get repaired. Either convert 30 days into minute fraction of time system is not available = (10/(30 × 24 × 60) × 100 = ·023%
Availability = 100 – 0.023 = 99.97%

44. Match each software lifecycle model in List-I to its description in List-II:

	List-I		List-II
A.	Code-and-Fix	**1.**	Assess risks at each step; do most critical action first.
B.	Evolutionary prototyping	**2.**	Build an initial small requirement specifications, code it, then "evolve" the specifications and code as needed.
C.	Spiral	**3.**	Build initial requirement specification for several releases, then design-and-code in sequence.
D.	Staged Delivery	**4.**	Standard phases (requirements, design, code, test) in order
E.	Waterfall	**5.**	Write some code, debug it, repeat (i.e. adhoc)

Codes:

	A	B	C	D	E
(a)	5	2	1	3	4
(b)	5	3	1	2	4
(c)	4	1	2	3	5
(d)	3	5	1	2	4

Ans : (a) (i) **Code and fix:** write some code, debug it repeat.
(ii) **Evaluation prototype:** system develop incrementally.
(iii) **Spiral model:** access risk at each step.

(iv) **Staged delivery:** is an evolutionary software release.
(v) **Water fall model:** sequential (non-iterative design process.

45. Match each software term in List-I to its description in List-II:

	List-I		List-II
A.	Wizards	**1.**	Forms that provide structure for a document
B.	Templates	**2.**	A series of commands grouped into a single command
C.	Macro	**3.**	A single program that incorporates most commonly used tools
D.	Integrated Software	**4.**	Step-by-step guides in application software
E.	Software Suite	**5.**	Bundled group of software programs

Codes:

	A	B	C	D	E
(a)	4	1	2	3	5
(b)	2	1	4	3	5
(c)	4	5	2	1	3
(d)	5	3	2	1	4

Ans : (a) Correct matching is:
(a)-(iv), (b)-(i), (c)-(ii), (d)-(iii), (e)-(v)

46. The ISO quality assurance standard that applies to software Engineering is

(a) ISO 9000: 2004
(b) ISO 9001: 2000
(c) ISO 9002: 2001
(d) ISO 9003: 2004

Ans : (b) The 9001 : 2000 is the quality assurance standard that applies to software engineering. The standard contains 20 requirement that must be present for an effective quality assurance system.

47. Which of the following are external qualities of a software product?

(a) Maintainability, reusability, portability, efficiency, correctness.

(b) Correctness, reliability, robustness, efficiency, usability.

(c) Portability, interoperability, maintainability, reusability.

(d) Robustness, efficiency, reliability, maintainability, reusability.

Ans : (b) External qualities of a software product:

- Correctness
- Reliability
- Robustness
- Efficiency
- Usability

Internal qualities of a software product

- Maintainability
- Reusability
- Portability
- Interoperability

48. Which of the following is/are CORRECT statement(s) about version and release?

I. A version is an instance of a system, which is functionally identical but non-functionally distinct from other instances of a system.

II. A version is an instance of a system, which is functionally distinct in some way from other system instances.

III. A release is an instance of a system, which is distributed to users outside of the development team.

IV. A release is an instance of a system, which is functionally identical but non-functionally distinct from other instances of a system.

(a) I and II (b) II and IV

(c) I and IV (d) II and III

Ans : (d) Version- An instance of a system which is functionally distinct in same way from other system instances.

Release- An instance of a system which is distributed to users outside of the development team.

49. The Unix Operating System Kernel maintains two key data structures related to processes, the process table and the user structure. Now, consider the following two statements:

I. The process table is resident all the time and contain information needed for all processes, even those that are not currently in memory.

II. The user structure is swapped or paged out when its associated process is not in memory, in order not to waste memory on information that is not needed.

Which of the following options is correct with reference to above statements?

(a) Only I is correct

(b) Only II is correct

(c) Both I and II are correct

(d) Both I and II are wrong

Ans : (c) Both statements are correct.

50. Consider a system which have 'n' number of processes and 'm' number of resources types. The time complexity of the safety algorithm, which checks whether a system is in safe state or not, is of the order of:

(a) $O(mn)$ (b) $O(m^2n^2)$

(c) $O(m^2n)$ (d) $O(mn^2)$

Ans : (d) Time complexity of banker's algorithm to deadlock avoidance is $O(mn^2)$.

51. An operating system supports a paged virtual memory, using a central processor with a cycle time of one microsecond. It costs an additional one microsecond to access a page other than the current one. Pages have 1000 words, and the paging device is a drum that rotates at 3000 revolutions per minute and transfers one million words per second. Further, one percent of all instructions executed accessed a page other than the current page. The instruction that accessed another page, 80% accessed a page already in memory and when a new page was required, the replaced page was modified 50% of the time. What is the effective access time on this system, assuming that the system is running only one process and the processor is idle during drum transfers?

(a) 30 microseconds (b) 34 microseconds

(c) 60 microseconds (d) 68 microseconds

Ans : (b) Effective access time

$= 0.99\ (1\ \mu\text{sec} + 0.008 \times 2\ \mu\text{sec}) + 0.002 \times (10000\ \mu\text{sec} + 1000\ \mu\text{sec}) + 0.001 \times (1000\ \mu\text{sec} + 1000\ \mu\text{sec})$

$= (0.99 + 0.016 + 22.0 + 11.0)\ \mu\text{sec}$

$= 34.0\ \mu\text{sec}$

52. Consider the following page reference string:

1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6

Which of the following options, gives the correct number of page faults related of LRU, FIFO, and optimal page replacement algorithms respectively, assuming 05 page frames and all frames are initially empty?

(a) 10, 14, 8

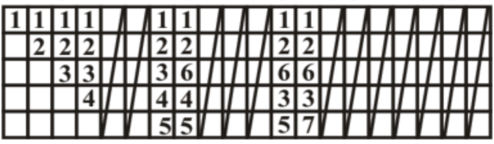
(b) 8, 10, 7

(c) 7, 10, 8

(d) 7, 10, 7

Ans : (b) (1) LRU algorithm:

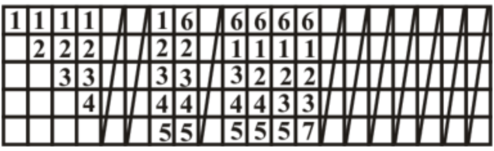
1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6



Total number of page fault = 8

(2) FIFO algorithm:

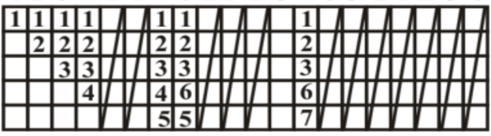
1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6



Total number of page fault = 10

(3) Optimal algorithm:

1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6



Total number of page fault = 7

53. Consider a file currently consisting of 50 blocks. Assume that the file control block and the index block is already in memory. If a block is added at the end (and the block information to be added is stored in memory), then how many disk I/O operations are required for indexed (single-level) allocation strategy?

(a) 1 (b) 101

(c) 27 (d) 0

Ans : (a) According to question file control block and the index block is already in memory. So, 1, 3 and 1, I/O disk operations are required for contiguous, linked and indexed allocation respectively.

54. An experimental file server is up 75% of the time and down for 25% of the time due to bugs. How many does this file server have to be replicated to give an availability of at least 99%?

(a) 2 (b) 4

(c) 8 (d) 16

Ans : (b) With K being the number of servers, we have $(1/4)^K < 0.01$, expressing the worst situation, when all servers are down, should happen at most (1/100) of the time.

This gives us $K = 4$.

55. Given the following two languages:

$$L_1 = \{u\,w\,w^R\,v \mid u, v, w \in \{a,b\}^+\}$$

$$L_2 = \{u\,w\,w^R\,v \mid u, v, w \in \{a,b\}^+, |u| \geq |v|\}$$

Which of the following is correct?

(a) L_1 is regular language and L_2 is not regular language.

(b) L_1 is not regular language and L_2 is regular language.

(c) Both L_1 and L_2 are regular languages.

(d) Both L_1 and L_2 are not regular languages.

Ans : (a) $L_1 = \{uww^R v | u, w, v \in \{a, b\}^+\}$ is regular because it can be described the regular expression

$L_1 = (a + b)^+ (aa + bb) (a + b)^+$

$L_2 = \{uww^R v | u, w, v \in \{a, b\}+, |u| \geq |v|\}$ is not regular, because we need a stack to compare length of $|u| \geq |v|$.

56. Given a Turing Machine

M = ({q_0, q_1}, {0, 1}, {0, 1, B}, δ, B, {q_1})

Where δ is a transition function defined as

$\delta(q_0, 0) = (q_0, 0, R)$

$\delta(q_0, B) = (q_1, B, R)$

The language L(M) accepted by turing machine is given as:

(a) 0* 1*
(b) 00*
(c) 10*
(d) 1*0*

Ans : (b) This Turing Machine accept 0*.
00* is subset of 0*. So, Turing Machine accepts all strings of 00*. The only string except 00* is ∈ accepted by this Turing Machine.

57. Let $G = (V, T, S, P)$ be a context-free grammar such that every one of its productions is of the form $A \rightarrow$ v, with |v| = k > 1. The derivation tree for any string $W \in L(G)$ has a height such that

(a) $h < \frac{(|W|-1)}{K-1}$

(b) $\log_k |W| \leq h$

(c) $\log_k |W| < h < \frac{(|W|-1)}{K-1}$

(d) $\log_k |W| \leq h \leq \frac{(|W|-1)}{K-1}$

Ans : (d) The derivation tree for any string w ∈ $L(G)$ has a height such that

$$\log_k |w| \leq h \leq \frac{(|w|-1)}{k-1}$$

58. Which of the following is not used in standard JPEG image compression?

(a) Huffman coding
(b) Runlength encoding
(c) Zig-zag scan
(d) K-L Transform

Ans : (d) K-L theorem is not used in image compression.

59. Which of the following is a source coding technique?

(a) Huffman coding (b) Arithmetic coding
(c) Run-length coding (d) DPCM

Ans : (d) DPCM is differential pulse code modulation which convert source analog into digital signal (i.e. source coding).

60. If the histogram of an image is clustered towards origin on X-axis of a histogram plot then it indicates that the image is

(a) Dark (b) Good contrast
(c) Bright (d) Very low contrast

Ans : (a) If the histogram of an image is clustered toward origin on x-axis of a histogram plot then it indicates that the image is dark.

If the histogram of an image is clustered towards origin on y-axis of a histogram plot then it indicates that the image is bight.

61. Consider the following linear programming problem:

Max, $z = 0.50x_2 - 0.10x_1$

Subject to the constraints

$$2x_1 + 5x_2 \leq 80$$
$$x_1 + x_2 \leq 20$$

and x_1, x_2

The total maximum profit (z) for the above problem is:

(a) 6 (b) 8
(c) 10 (d) 12

Ans : (b) when,

$x_2 = 16$ and $x_2 = 0$

then Max $z = 8$

62. Consider the following statements:

(i) If primal (dual) problem has a finite optimal solution, then its dual (primal) problem has a finite optimal solution.

(ii) If primal (dual) problem has an unbounded optimum solution, then its dual (primal) has no feasible solution at all.

(iii) Both primal and dual problems may be infeasible.

Which of the following is correct?

(a) (i) and (ii) only (b) (i) and (iii) only
(c) (ii) and (iii) only (d) (i), (ii) and (iii)

Ans : (d) These are fundamental duality theorem. All options are correct.

63. Consider the following statements:

(i) Assignment problem can be used to minimize the cost.

(ii) Assignment problem is a special case of transportation problem.

(iii) Assignment problem requires that only one activity be assigned to each resource.

Which of the following options is correct?

(a) (i) and (ii) only (b) (i) and (iii) only

(c) (ii) and (iii) only (d) (i), (ii) and (iii)

Ans : (d) Assignment problem is special case of transportation problem. It requires that only one activity be assigned to each resource and it can be used to minimize the cost. All statements are correct.

64. What are the following sequence of steps taken in designing a fuzzy logic machine?

(a) Fuzzification $\rightarrow$ Rule evaluation $\rightarrow$ Defuzzification

(b) Fuzzification $\rightarrow$ Defuzzification $\rightarrow$ Rule evaluation

(c) Rule evaluation $\rightarrow$ Fuzzification $\rightarrow$ Defuzzification

(d) Rule evaluation $\rightarrow$ Defuzzification $\rightarrow$ Fuzzification

Ans : (a) Fuzzification $\rightarrow$ Rule evaluation $\rightarrow$ Defuzzification

The process of fuzzy reasoning is incorporated into what is called a Fuzzy Inferening system.

It involves three steps –

1. Fuzzification
2. Rule evaluation
3. Defuzzification

65. Which of the following 2 input Boolean logic functions is linearly inseparable?

(i) AND **(ii) OR**

(iii) NOR **(iv)XOR**

(v) NOTXOR

(a) (i) and (ii) (b) (ii) and (iii)

(c) (iii), (iv) and (v) (d) (iv) and (v)

Ans : (d) XOR is linearly inseparable, if XOR is, then swapping the labels on the points will not alter the structure of the space, and NOT XOR is just XOR with the 1's swapped with 0's and vice-versa.

66. Let *R* and *S* be two fuzzy relations defined as

$$R = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\!\!\begin{array}{c} \begin{array}{cc} y_1 & y_2 \end{array} \\ \begin{bmatrix} 0.7 & 0.5 \\ 0.8 & 0.4 \end{bmatrix} \end{array} \text{ and } S = \begin{array}{c} \\ y_1 \\ y_2 \end{array}\!\!\begin{array}{c} \begin{array}{ccc} z_1 & z_2 & z_3 \end{array} \\ \begin{bmatrix} 0.9 & 0.6 & 0.2 \\ 0.1 & 0.7 & 0.5 \end{bmatrix} \end{array}$$

Then, the resulting relation, *T*, which relates elements of universe of *X* to elements of universe of *Z* using max-product composition is given by

(a) $$T = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\!\!\begin{array}{c} \begin{array}{ccc} z_1 & z_2 & z_3 \end{array} \\ \begin{bmatrix} 0.68 & 0.89 & 0.39 \\ 0.76 & 0.72 & 0.32 \end{bmatrix} \end{array}$$

(b) $T = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{array}{c} \begin{array}{ccc} z_1 & z_2 & z_3 \end{array} \\ \begin{bmatrix} 0.68 & 0.89 & 0.39 \\ 0.72 & 0.76 & 0.32 \end{bmatrix} \end{array}$

(c) $T = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{array}{c} \begin{array}{ccc} z_1 & z_2 & z_3 \end{array} \\ \begin{bmatrix} 0.63 & 0.42 & 0.35 \\ 0.72 & 0.48 & 0.20 \end{bmatrix} \end{array}$

(d) $T = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{array}{c} \begin{array}{ccc} z_1 & z_2 & z_3 \end{array} \\ \begin{bmatrix} 0.05 & 0.35 & 0.14 \\ 0.04 & 0.28 & 0.16 \end{bmatrix} \end{array}$

Ans : (c) By **max-min composition:**

$\mu_T(x_1, z_1) = \max(\min(0.7, 0.9), \min(0.5, 0.1))$

$= 0.7$

and the rest

$T = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{array}{c} \begin{array}{ccc} z_1 & z_2 & z_3 \end{array} \\ \begin{bmatrix} 0.7 & 0.6 & 0.5 \\ 0.8 & 0.6 & 0.4 \end{bmatrix} \end{array}$

and by max-product composition:

$\mu_T(x_2, z_2) = \max((0.8, 0.6), (0.4, 0.7)) = 0.48$

$T = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{array}{c} \begin{array}{ccc} z_1 & z_2 & z_3 \end{array} \\ \begin{bmatrix} 0.63 & 0.42 & 0.35 \\ 0.72 & 0.48 & 0.20 \end{bmatrix} \end{array}$

67. Consider the following operations to be performed in Unix:

"The pipe sorts all files in the current directory modified in the month of "June" by order of size and prints them to the terminal screen. The sort option skips ten fields then sorts the lines in numeric order."

Which of the following Unix command will perform above set of operation?

(a) Is - *l* |grep "June" | sort + 10n

(b) Is - *l* |grep "June" | sort + 10r

(c) Is - *l* |grep -v "June" | sort + 10n

(d) Is - *l* |grep -n "June" | sort + 10x

Ans : (a) Correct unix command is - Is - *l* |grep "June" | sort + 10n.

68. Which of the following statements is incorrect for a Windows Multiple Document Interface (MDI)?

(a) Each document in an MDI application is displayed in a separate child window within the client area of the application's main window.

(b) An MDI application has three kinds of windows namely a frame window, an MDI client window and number of child windows.

(c) An MDI application can support more than one kind of document.

(d) An MDI application displays output in the client area of the frame window.

Ans : (d) It displays the MDI client window. An MDI application does not display output in the client areas of the frame window.

69. Which of the following statement(s) is/are True regarding 'nice' command of UNIX?

I. It is used to set or change the priority of a process.

II. A process's nice value can be set at the time of creation.

III. 'nice' takes a command line as an argument.

(a) I, II only (b) II, III only

(c) I, II, III (d) I, III only

Ans : (c) A process's nice value can be set at the time of creation with nice command and adjusted later with renice command.

(i) To set or change the priority of a process.

(ii) It takes a command line as an argument.

(iii) It's value can be set at the time of creation.

70. Let v(x) mean x is a vegetarian, m(y) for y is meat, and e(x, y) for x eats y. Based on these, consider the following sentences:

I. $\forall x v(x) \Leftrightarrow (\forall y e(x, y) \Rightarrow \neg m(y))$

II. $\forall x v(x) \Leftrightarrow (\neg(\exists y\, m(y) \wedge e(x, y)))$

III. $\forall x v(\exists y\, m(y) \wedge e(x, y)) \Leftrightarrow \neg v(x)$

One can determine that

(a) Only I and II are equivalent sentences

(b) Only II and III are equivalent sentences.

(c) Only I and III are equivalent sentence.

(d) I, II, and III are equivalent sentences.

Ans : (d) $= \forall x u(x) \Leftrightarrow (\forall y e(x, y) \Rightarrow \neg m(y))$

$= \forall x u(x) \Leftrightarrow (\neg(\exists y\, m(y) \wedge e(x, y)))$

$= \forall x(\exists y\, m(y) \wedge e(x, y)) \Leftrightarrow \neg u(x)$

$= \forall x(\neg(\forall y\, m(y) \Rightarrow e(x, y)) \Leftrightarrow \neg u(x)$

So, all statements are equivalent.

71. Match each Artificial Intelligence term in List-I that best describes a given situation in List-II:

	List-I		List-II
A.	Semantic Network	**1.**	Knowledge about what to do as opposed to how to do it.
B.	Frame	**2.**	A premise of a rule that is not concluded by any rule.

C.	Declarative knowledge	**3.**	A method of knowledge representation that uses a graph.
D.	Primitive	**4.**	A data structure representing stereotypical knowledge.

Codes:

	A	B	C	D
(a)	4	1	2	3
(b)	4	3	1	2
(c)	4	3	2	1
(d)	3	4	1	2

Ans : (d) A **semantic network** represents semantic relations between concepts. This is often used as a form of **knowledge representation.**

A **frame** is a data-structure for representing a stereotyped situation, like being in a certain kind of living room.

Declarative knowledge refers to factual knowledge and information that a person knows.

A **procedural knowledge,** on the other hand, is knowing how to perform certain activities.

Primitive is premise of a rule that is not concluded by any rule.

72. In Artificial Intelligence, a semantic network

(a) is a graph-based method of knowledge representation where nodes represent concepts and arcs represent relations between concepts.

(b) is a graph-based method of knowledge representation where nodes represent relations between concepts and arcs represent concepts.

(c) represents an entity as a set of slots and associated rules.

(d) is a subset of first-order logic.

Ans : (a) In Artificial Intelligence is a graph-based method of knowledge representation where nodes represent concepts and arcs represent relations between concepts. In other words Semantic Network representation nodes represents concepts, instances or events and link represents relationship between nodes, such as is-a, part-of, kind-of, interactive relations.

73. Criticism free idea generation is a factor of _________.

(a) Decision Support System
(b) Group Decision Support System
(c) Enterprise Resource Support System
(d) Artificial Intelligence

Ans : (b) Criticism free idea generation is a factor of group decision support system.

74. Consider the following logical inferences:

I_1: If it is Sunday then school will not open.
The school was open.
Inference: It was not Sunday.

I_2: If it is Sunday then school will not open.
It was not Sunday.
Inference: The school was open.

Which of the following is correct?

(a) Both I_1 and I_2 are correct inferences.
(b) I_1 is correct but I_2 is not a correct inference.
(c) I_1 is not correct but I_2 is a correct inference.
(d) Both I_1 and I_2 are not correct inferences.

Ans : (b) Let p : It is Sunday
q : School will not open

Then

$$I_1: \quad \begin{array}{l} p \to q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

It is true, since contrapositive is also true.

$= ((p \to q) \wedge \neg p) \to \neg p$

$= (\neg(\neg p \vee q) \vee q) \vee \neg p$

$= p \wedge \neg q \vee q \vee \neg p$

$= p \vee q \neg p$ = True (tautology)

$$I_2: \quad \begin{array}{l} p \to q \\ \neg p \\ \hline \therefore \neg q \end{array}$$

$= ((p \to q) \wedge \neg p) \to q$

$= (\neg(\neg p \vee q) \vee p) \vee \neg q$

$= (p \wedge \neg q) \vee p \vee \neg q$

$= p \wedge \neg q$ = Not tautology but contingency.

75. Which formal system provides the semantic foundation for Prolog?

(a) Predicate calculus
(b) Lambda calculus
(c) Hoare logic
(d) Propositional logic

Ans : (a) Pure prolog is based on a subset of first-order predicate logic, horn clauses which is Turing Complete.